ModelIt!

Student Activity Pack

Why Do Environmental Inequities Exist?

Modeling Pollution Exposure Disparities, Environmental Justice, and Systemic Solutions

Grade 12L2 | NGSS HS-ESS3-3 | Systems and System Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Do you think all communities in your city or region have the same air quality, water quality, and access to green space? Why or why not?**

2. **Why do you think pollution sources like factories, highways, and waste facilities tend to be located in certain neighborhoods rather than others? What determines where they are placed?**

3. **What does the term environmental justice mean to you? Can you give an example of environmental injustice?**

4. **If a community has higher pollution and worse health outcomes than average, whose responsibility is it to fix the problem — the community itself, the polluting companies, or the government?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Pollution Source Density
- Community Socioeconomic Status
- Pollution Exposure Level
- Health Outcome Disparity
- Regulatory Enforcement Level
- Community Political Power
- Green Infrastructure Access

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Status Quo

Pollution Sources: Concentrated in Low-Income Areas | Enforcement: Weak in Low-Income Areas | Political Power: Correlated with Income

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Equal Enforcement

Pollution Sources: Existing Distribution | Enforcement: Equal Across All Areas | Political Power: Current

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Systemic Transformation

Pollution Sources: Reduced in Overburdened Areas | Enforcement: Equal + Enhanced | Political Power: Equalized | Green Infrastructure: Invested

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Historical Zoning and Redlining Legacy — The lasting impact of discriminatory zoning and lending practices from the 20th century — neighborhoods that were redlined by the Home Owners\' Loan Corporation in the 1930s are, on average, 5 degrees Fahrenheit hotter, have significantly less tree canopy, and host more pollution sources today than neighborhoods that were rated favorably, demonstrating how historical policy created lasting environmental inequity

Green Gentrification Risk — The paradoxical process by which environmental improvements in disadvantaged communities increase property values and attract wealthier residents, displacing the original community members who fought for and deserve the benefits — anti-displacement policies like community land trusts and right-to-return provisions are essential safeguards

Cumulative Impact Screening Tools — Data platforms like CalEnviroScreen and EJScreen that map the combined environmental, health, and socioeconomic burdens on communities — these tools enable data-driven identification of the most overburdened communities and provide the evidence base for targeted investment and enforcement

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If clean air and clean water are basic human needs, why do low-income communities and communities of color consistently bear the greatest burden of pollution, toxic waste, and environmental hazards — and what systemic forces created and maintain these patterns of environmental inequity? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that environmental inequity is the result of systemic patterns rather than random chance? What specific feedback loops maintain the disparity?**

2. **Why did your model show that equal enforcement alone is insufficient to achieve environmental justice? What structural factors persist even with equal enforcement?**

3. **What makes Community Political Power the strongest leverage point in your model? How does empowerment break feedback loops that other interventions do not address?**

4. **How does the concept of cumulative impact explain why a community surrounded by multiple pollution sources faces health effects far worse than what any single source would cause?**

5. **What is the green gentrification problem, and how can environmental justice plans prevent environmental improvements from displacing the communities they are meant to benefit?**

8 STEM Challenge: Design an Environmental Justice Action Plan

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YOUR ENGINEERING MISSION

Design a comprehensive environmental justice plan for an overburdened community that addresses pollution reduction, health disparities, community empowerment, and equitable development using your model evidence about systemic drivers of environmental inequity.

THE SCENARIO

An urban neighborhood of 30,000 people — 85% low-income, 92% people of color — is surrounded by a petroleum refinery, two chemical plants, a major highway interchange, and a waste transfer station. Asthma rates are 4 times the national average, cancer rates are elevated, and life expectancy is 12 years shorter than the affluent suburb 5 miles away. Environmental violations by nearby facilities are rarely enforced. Residents have organized a community environmental justice group and secured a $50 million investment from the state. Your team must design a plan that uses this investment to achieve maximum environmental and health improvement, addressing both immediate pollution exposure and the systemic factors that created the inequity.

Engineering Design Thinking

1. WHAT SPECIFIC POLLUTION REDUCTION MEASURES DOES YOUR PLAN IMPLEMENT FOR EACH SOURCE (REFINERY, CHEMICAL PLANTS, HIGHWAY, WASTE STATION), AND WHAT EXPOSURE REDUCTION DO YOU PROJECT?

2. HOW DOES YOUR PLAN BUILD PERMANENT COMMUNITY POLITICAL POWER TO PREVENT FUTURE ENVIRONMENTAL INEQUITY — NOT JUST ADDRESS THE CURRENT SITUATION?

3. WHAT GREEN INFRASTRUCTURE INVESTMENTS DOES YOUR PLAN INCLUDE, AND HOW DO THEY ADDRESS BOTH ENVIRONMENTAL QUALITY AND COMMUNITY HEALTH?

4. HOW DOES YOUR PLAN ENSURE THAT ENVIRONMENTAL IMPROVEMENTS DO NOT TRIGGER GENTRIFICATION THAT DISPLACES THE COMMUNITY MEMBERS WHO FOUGHT FOR AND DESERVE THE BENEFITS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Environmental justice advocates and community organizers who fight for equitable environmental protection work for grassroots organizations, legal aid societies, and government agencies, earning $45,000-$95,000/year. Environmental health scientists who study the relationship between pollution exposure and health outcomes in affected communities earn $60,000-$125,000/year. Environmental lawyers who bring legal action against polluters and advocate for regulatory reform earn $65,000-$160,000/year at public interest firms, government agencies, and private practice.